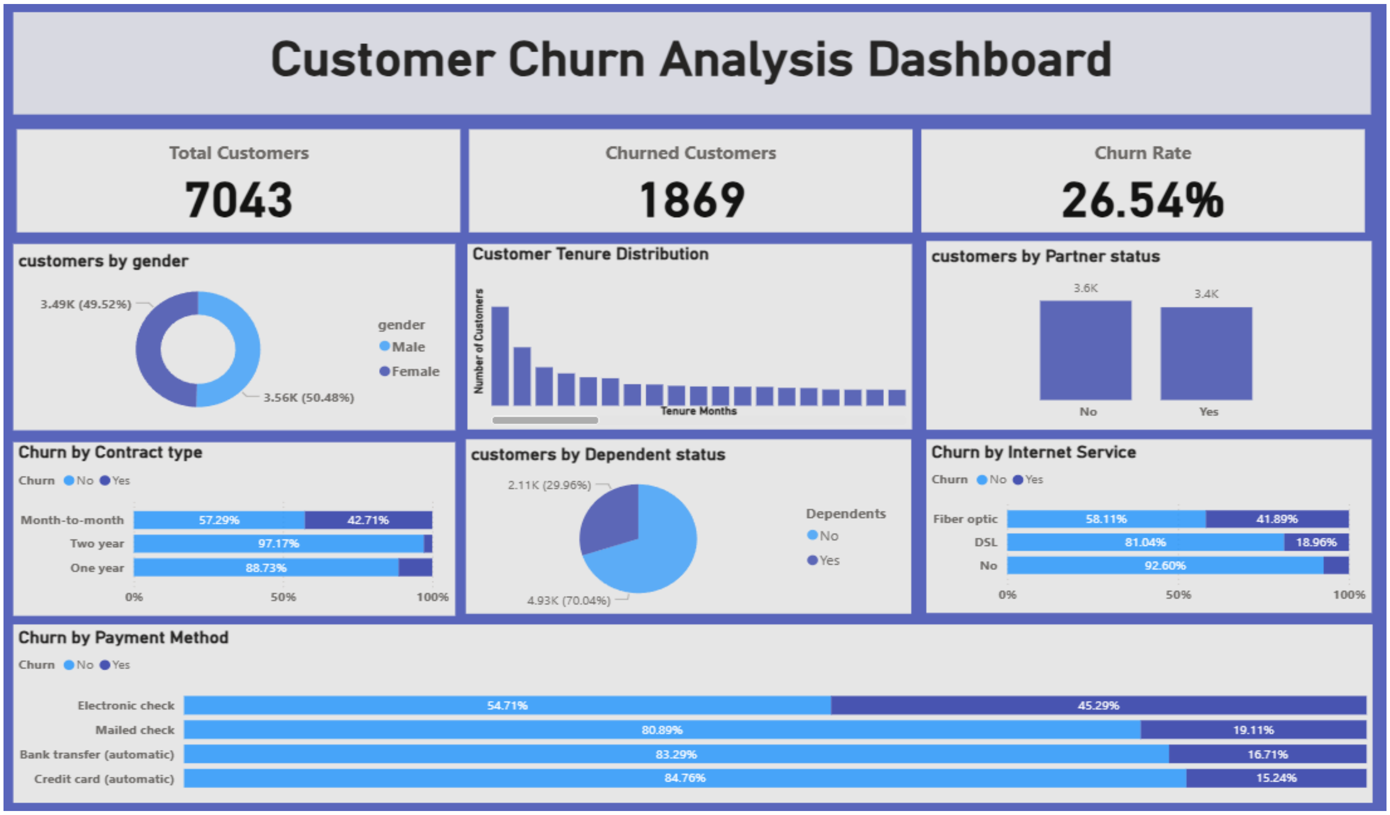
Customer Churn Analysis Dashboard Report



1. Executive Summary

This project focuses on analyzing customer churn behavior within a telecommunications company using Power BI. The dashboard was developed to identify customer retention patterns, understand churn-related factors, and provide business recommendations to reduce customer loss. The analysis helps stakeholders understand how customer demographics, contract types, internet services, payment methods, and tenure influence customer churn.

2. Business Problem Statement

Customer churn directly affects the profitability and long-term sustainability of telecom companies. High churn rates result in revenue loss, increased customer acquisition costs, and reduced business growth. The objective of this project is to identify the major factors contributing to customer churn and create an interactive Power BI dashboard for business decision-making.

3. Project Objectives

• Analyze customer churn trends and patterns.
• Identify high-risk customer groups.
• Create interactive visual dashboards.
• Analyze churn based on demographics and services.
• Build KPI-driven reporting solutions.
• Generate actionable business recommendations.

4. Dataset Overview

The dataset contains customer-level information from a telecommunications company. It includes demographic details, internet services, contract information, payment methods, tenure, and churn status. Important dataset attributes include:

• Customer ID
• Gender
• Partner Status
• Dependents
• Internet Service
• Contract Type
• Monthly Charges
• Payment Method
• Customer Tenure
• Churn Status

5. Tools and Technologies Used

The project was developed using the following technologies:

• Power BI – Dashboard development and business intelligence reporting.
• Power Query – Data cleaning and transformation.
• DAX – KPI calculations and analytical measures.
• CSV Dataset – Source data storage.
• Microsoft Power BI Desktop – Visualization platform.

6. Data Cleaning and Transformation

Data preprocessing was performed to improve data quality and ensure analytical accuracy. Major preprocessing activities included:

• Handling missing values.
• Standardizing categorical fields.
• Validating data consistency.
• Removing formatting inconsistencies.
• Creating calculated columns and measures.
• Optimizing data types for reporting.

7. Dashboard KPIs

The dashboard contains key performance indicators to summarize customer churn performance.

• Total Customers: 7043
• Churned Customers: 1869
• Churn Rate: 26.54%

8. Dashboard Components

The Power BI dashboard contains several visualizations and analytical components:

Customer Demographics
Analyzes customer distribution by gender, partner status, and dependent status.

Customer Tenure Distribution
Visualizes how long customers remain associated with the company.

Churn by Contract Type
Compares churn rates between month-to-month, one-year, and two-year contracts.

Churn by Internet Service
Analyzes churn based on Fiber Optic, DSL, and no internet services.

Churn by Payment Method
Identifies churn patterns based on customer payment preferences.

9. Key Findings and Insights

The project identified several important business insights:

• Month-to-month customers exhibit the highest churn rates.
• Customers using Fiber Optic services show higher churn percentages.
• Electronic check payment users are more likely to churn.
• Long-term contracts significantly reduce churn probability.
• Customer tenure strongly influences customer retention.

10. Business Recommendations

Based on the analysis, the following recommendations were proposed:

• Promote long-term contract subscriptions.
• Improve Fiber Optic customer support and service quality.
• Encourage customers to use automatic payment methods.
• Offer loyalty rewards for long-tenure customers.
• Create targeted retention campaigns for high-risk customers.

11. Power BI Skills Demonstrated

This project demonstrates practical business intelligence and dashboard development skills including:

• Data Cleaning and Transformation
• DAX Calculations
• KPI Development
• Dashboard Design
• Interactive Visualizations
• Business Analytics Reporting
• Customer Retention Analysis

12. Challenges Faced

Several practical challenges were encountered during the project:

• Handling missing or inconsistent values.
• Designing balanced visual layouts.
• Optimizing dashboard readability.
• Managing data relationships and DAX calculations.
• Ensuring accurate KPI representation.

13. Learning Outcomes

This project enhanced practical skills in:

• Business Intelligence
• Power BI Dashboard Development
• Customer Analytics
• Data Transformation
• Data Visualization
• Analytical Thinking
• KPI Reporting

14. Future Improvements

Future enhancements for this project may include:

• Machine Learning-based churn prediction.
• Real-time customer monitoring dashboards.
• AI-driven retention recommendations.
• Advanced customer segmentation.
• Predictive analytics integration.

15. Conclusion

The Customer Churn Analysis Dashboard successfully demonstrates the use of Power BI and business intelligence techniques to analyze customer behavior and identify churn trends. The project provides valuable insights into customer retention strategies and supports data-driven business decision-making within the telecommunications industry.

16. Author

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Dashboard KPI Summary

KPI	Value
Total Customers	7043
Churned Customers	1869

Churn Rate	26.54%